Ollscoil Teicneolaíochta an Atlantaigh

Atlantic Technological University

Practical No: **2**

Date: **28th September 2026**

Title: **Transistor Sampling**

Name: **David Roberto Tapia Caicedo**

Course: **Bachelor of Engineering in Electronic Engineering**

Subject: **Digital Communications and Transmission**

Lecturer: **Ben Graham**

Summary:

In this lab the sampling and reconstruction of an analogue sine wave will be investigated using a transistor and a low-pass filter and practically verifying the Nyquist sampling theorem.

Objective:

The objective of this laboratory experiment was to learn and evaluate about sampling and reconstruction circuits to practically verify the Nyquist sampling theorem. Additionally, verify how different sampling rates affect the wave and compare a standard BC547 transistor with a CD4066 analogue switch.

COMPONENTS AND EQUIPMENTS REQUIRED:

BC547 NPN Transistor
CD 4066
Resistors – 8.2K Ω, 1K Ω ,470 Ω and 220 Ω
Capacitor- 1µF
Signal Generators.
Oscilloscope
Bread Board
Wires and probes

Theory:

The process of converting a continuous analogue signal into a digital format generally consists of three main stages: sampling, quantization, and encoding. However, the main and most critical step is sampling, where the continuous amplitude of the analogue waveform is captured at regular, discrete time intervals. According to the Nyquist Sampling Theorem, this sampling rate must be strictly greater than twice the maximum frequency of the input signal to allow for accurate recovery and avoid aliasing.

In this practical experiment, a basic sampling mechanism is implemented using a transistor acting as an analogue switch. Driven by a square wave clock pulse, the transistor chops the continuous input sine wave into discrete amplitude pulses. Subsequently, at the receiver stage, these discrete samples must be transformed back into a continuous analogue format. The reconstruction circuit operates as a low-pass filter with a cut-off frequency designed to match that of the analogue input signal.

Procedure:

The experiment was initiated by learning the basics about CircuitMaker because there was no previous knowledge. Simulating the sampling and reconstruction circuit in this program to analyze theoretical waveform behaviour before to physical assembly. Following the schematic, the circuit was constructed on a breadboard using a BC547 NPN bipolar junction transistor, connected with appropriate collector and resistors.

Two independent signal generators were calibrated and connected to the setup. The first generator supplied the analogue message signal, configured as a 20 Hz sinusoidal wave with an amplitude of 2.5 V. The second one provided the sampling clock, delivering a square wave with a 50% duty cycle to the base of the transistor. The output of the transistor switch was routed into a passive first-order low-pass filter consisting of an 8.2 kΩ resistor, R3, and a 1 µF capacitor, C1, to reconstruct the continuous signal.

Oscilloscope probes were positioned at key test points across the circuit to simultaneously observe the analogue input, the sampling pulses, the chopped collector output, and the filtered output waveform. To evaluate the Nyquist sampling criterion, the sampling frequency was intentionally adjusted below and above the 250 Hz threshold, and the resulting waveform distortion and envelope reconstruction were recorded. Oscilloscope cursors were utilized to measure the horizontal time delay between the input sine wave and the reconstructed signal to determine the practical phase shift. Finally, the discrete BC547 BJT transistor was replaced by a CD4066 CMOS bilateral analogue switch under identical input and filtering conditions to compare signal fidelity, switching linearity, and the presence of DC offset voltages.

Results

The diagram has been recreated in CircuitMaker, and all waveforms have been analyzed. The wave was sampled using 1 ms wide pulses, as can be observed in Figure 4. Upon passing the wave through the DAC, we can see that the resulting wave (see Figure 5) is like the initial analogue wave (see Figure 2), although with slight alterations due to the components not being ideal.

Circuit Diagram:

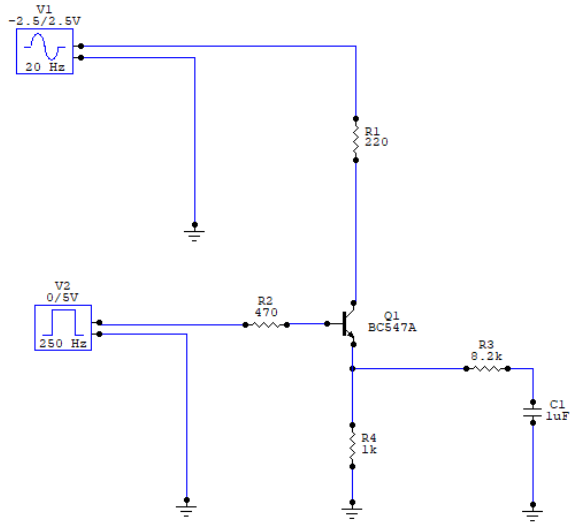


Figure 1: Diagram in CircuitMaker

Waveforms:

Analogue input signal:

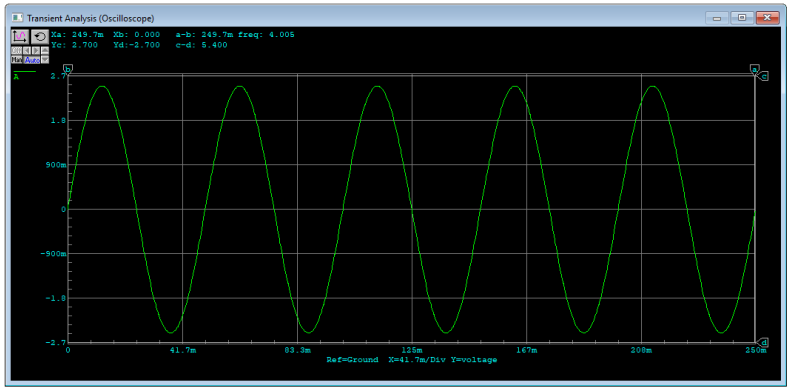


Figure 2: Analogue input signal wave

Sampling signal:

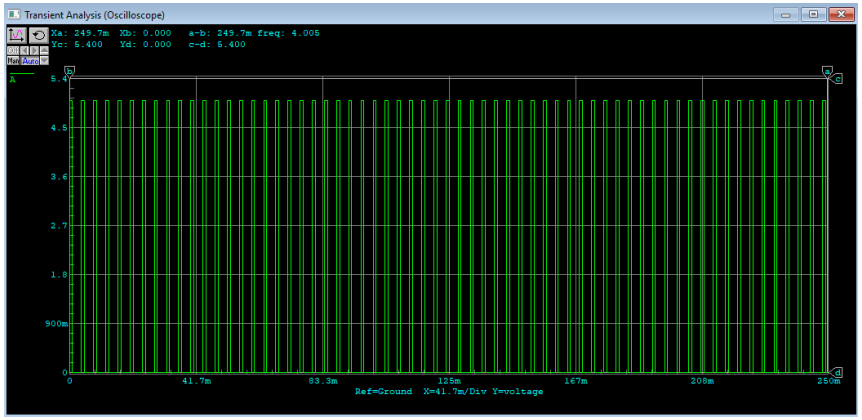


Figure 3: Sampling signal wave

Sampled output:

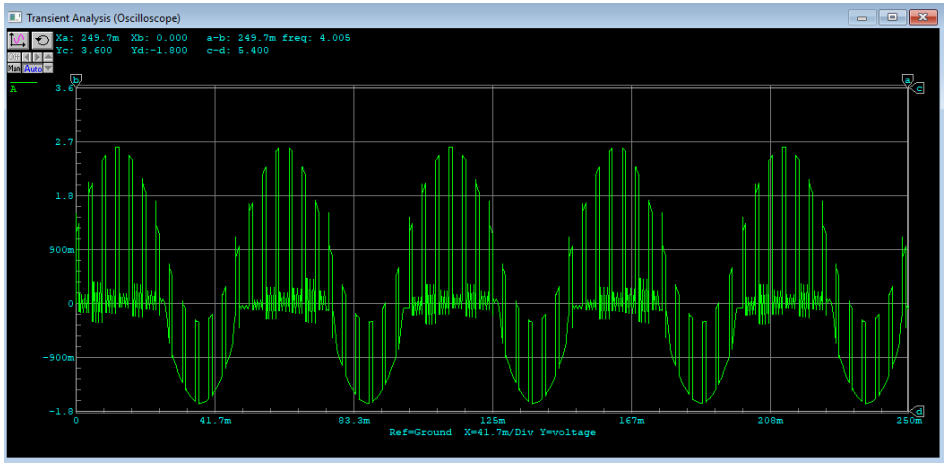


Figure 4: Samplet output wave

Reconstructed Signal:

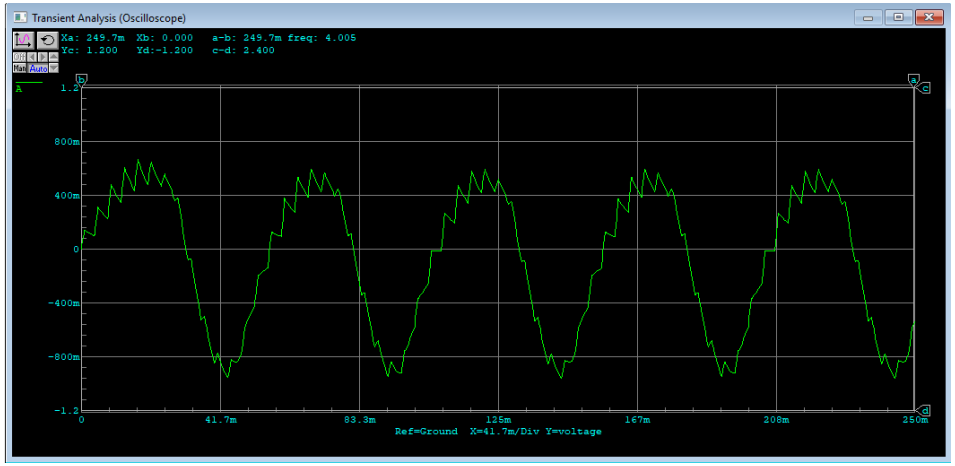


Figure 5: Reconstructed Wave

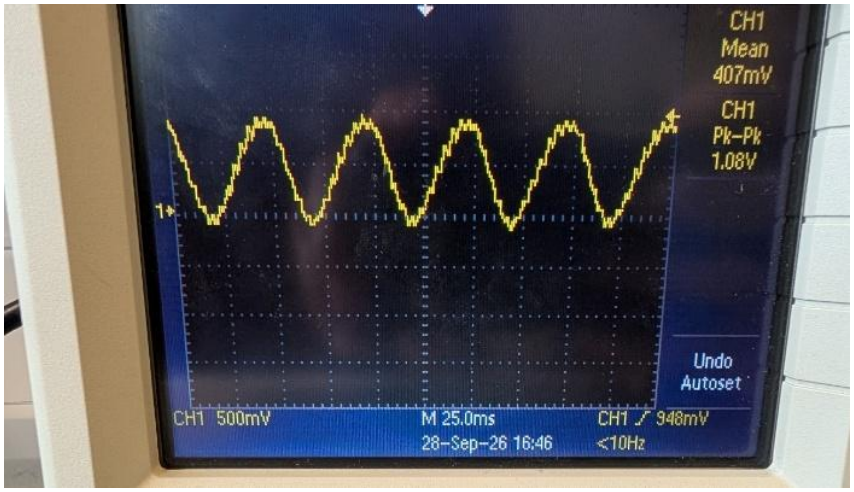


Figure 6: Reconstructed Wave on the Oscilloscope

Frequency above and below 250 Hz and Understanding of Nyquist sampling rate

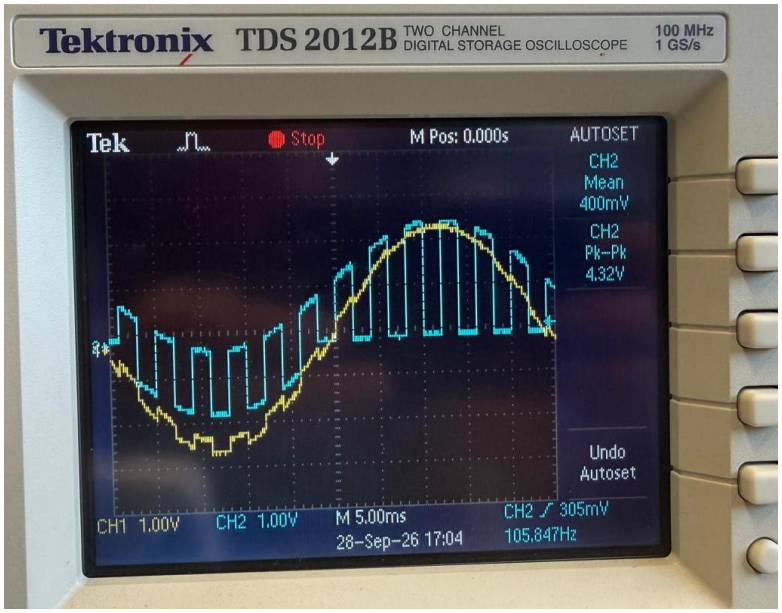


Figure 7: Signals with frequencies above and below 250Hz

The Nyquist sampling rate dictated that for the correct sampling of a signal, the sampling frequency had to be at least twice the highest frequency of the signal. This was to ensure that all necessary information was captured and to avoid the error of aliasing, in which the frequencies overlapped. This could be observed when the signal was tested below 250 Hz, where fewer and wider samples were verified per cycle, failing to adequately track the shape of the original 20 Hz sine wave. Consequently, the filter could not reconstruct the initial waveform, demonstrating that once aliasing occurred, the initial signal could not be recovered.

Estimate the phase change between the original waveform and the reconstructed waveform

To perform the theoretical calculation, we will first calculate X_C. To do this, we will use resistor R3 and capacitor C1, and discover Theorical value = 44.1º as shown in Figure 7.

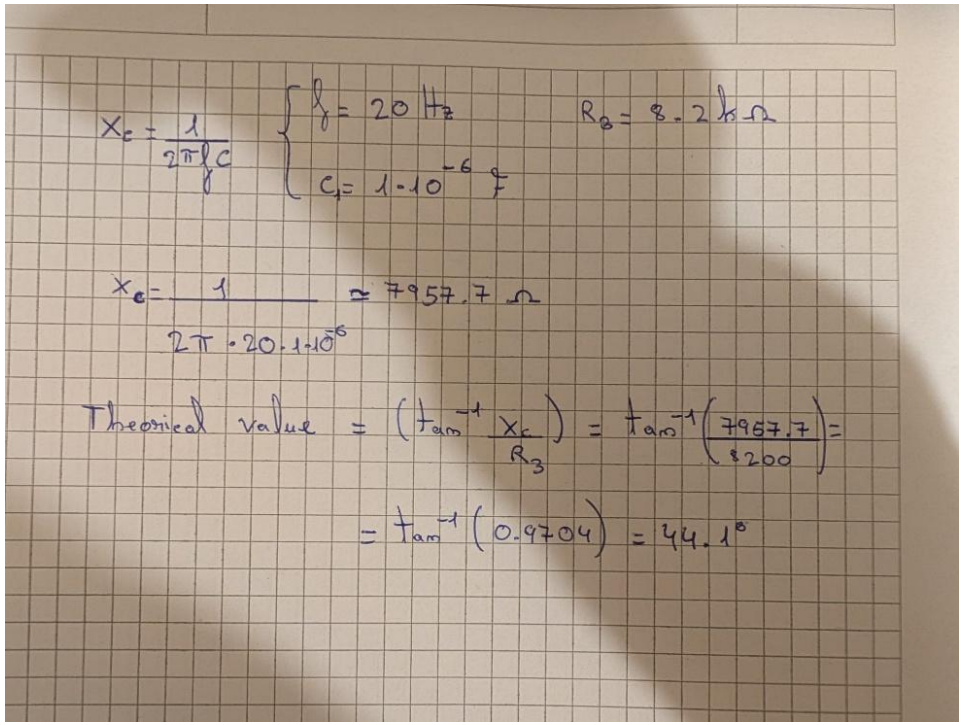


Figure 8: Calculation of the theorical value

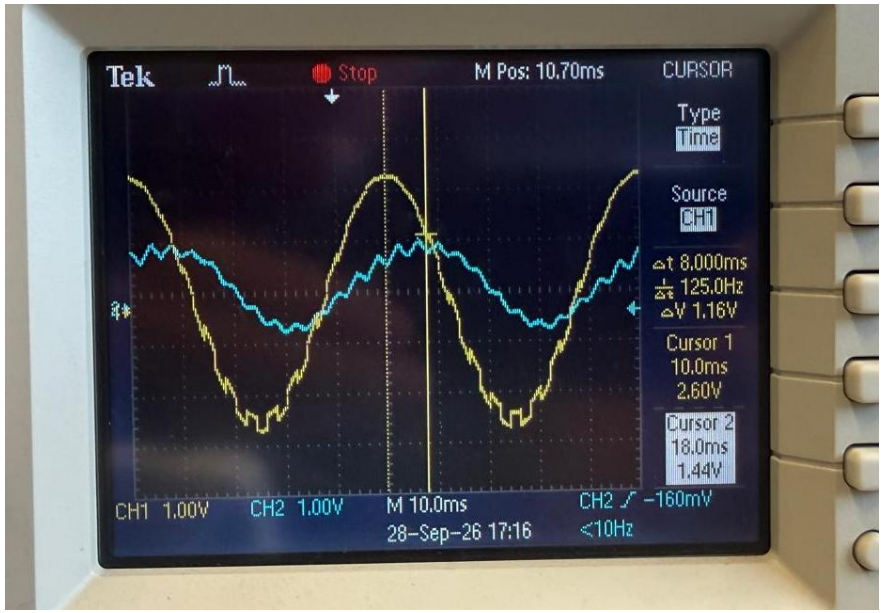


Figure 9: Measured time delay

Measured time delay (Δt) via oscilloscope cursors = 8.0 ms
T = 1/f = 1/20Hz = 50 ms
Practical phase = (8.0 ms/50 ms) * 360º = 57.6º
From these results, we can see a reasonably notable difference between the calculated theoretical value, 44.1°, and the actual value observed with the oscilloscope, 57.6°. This value is derived from the 8 ms delay between the peaks of the original and the reconstructed waveforms. This 13.5° difference between the two signals may be due to component tolerances, which are not ideal. We can also see that the BC547 BJT does not operate as an ideal switch either, since it introduces a voltage drop, its non-linear and it also allows small amount of current to discharge when switch is off.

Frequency of the reconstruction circuit

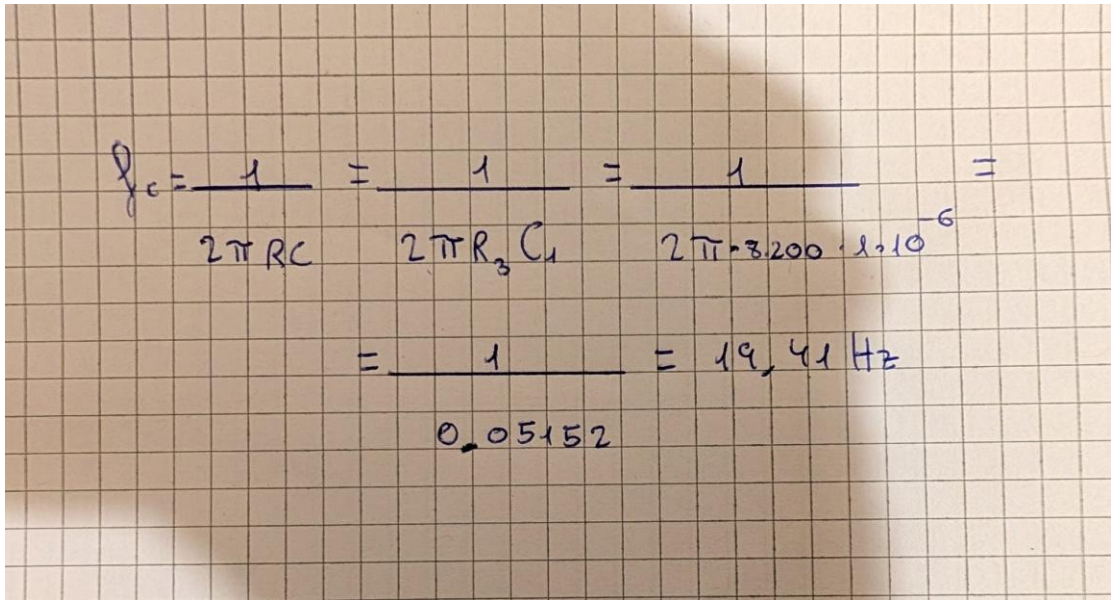


Figure 10: Calculation of the cut off frequency of the reconstruction circuit

The value, 19.41 Hz, is highly significant because the original analogue sine wave applied to the circuit has a frequency of 20 Hz. This means the signal was operating just above the filter's cut-off point, right on the edge of the passband. Because of this, the filter naturally began to attenuate and delay the signal, which explains why we observed a drop in amplitude and a phase shift in the reconstructed wave during the experiment.

Signals using CD4066 transistor

To improve the sampling accuracy, the BC547 was replaced with a CD4066 CMOS switch. The analogue signal was fed into the switch input, its output went to the low-pass reconstruction filter (R_3 and C_1). On the oscilloscope the sampled

waveform showed accurate transitions and a much cleaner waveform than the transistor circuit, note the less precise waveform below the x-axis in Figure 10. Finally, the reconstructed signal tracked the original 20 Hz sine wave closely with minimal amplitude distortion.



Figure 11: Sampled output Signal with BC547 NPN Transistor



Figure 12: Sample Output Diagram with CD4066 transistor

Conclusion:

In this practical, the processes of analogue sampling and signal reconstruction were evaluated, practically validating the Nyquist Sampling Theorem.

Through circuit simulation and experimental assembly, sampling was proved to be an indispensable stage in modern communication systems. It enables the sampling of continuous real-world phenomena for subsequent quantization and encoding, allowing for efficient data storage, robust transmission against electromagnetic noise, and time-division multiplexing.

Varying the sampling frequency above and below the 250 Hz threshold highlighted the direct consequences of the Nyquist rate. While an adequate sampling rate allowed the sinusoidal envelope to be reconstructed with sufficient resolution, reducing the frequency led to an irreversible loss of information due to aliasing, making it impossible for the reconstruction filter to recover the original 20 Hz waveform.

Furthermore, testing the RC filter also showed some practical limits. The calculated cut-off frequency was 19.41 Hz, which is almost the same as our 20 Hz input signal. Because of this, the filter reduced the amplitude and caused an 8.0 ms delay (a phase shift of 57.6° in practice, compared to the 44.1° theoretical value).

Finally using the CD4066 chip instead of the BC547 transistor gave much better results. The CMOS switch improved the linearity of the samples because it doesn't suffer from the saturation voltage drops or leakage currents that the BJT has.